Ayush Panda

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PROFESSIONAL SUMMARY

5th-semester B.Tech CSE student at VSSUT (CGPA 8.27) who ships full-stack and AI products end-to-end. Built 4 live SaaS apps and advanced auth systems spanning OAuth 2.0/OIDC, RBAC, Zanzibar-style authorization, Redis rate limiting, Kafka event streaming, and observability. Open to software engineering internships.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, HTML, CSS, SQL
Frontend: React, Next.js, Tailwind CSS, TanStack Router, Framer Motion
Backend: Node.js, Express, tRPC, REST, Socket.io, Server-Sent Events (SSE)
Auth & Security: OAuth 2.0, OpenID Connect (OIDC), PKCE, JWKS, JWT, RBAC, relationship-tuple authorization, rate limiting, distributed locks
Data: PostgreSQL, MongoDB, Redis, Firebase, Drizzle ORM, Prisma, pgvector
Infra & Ops: Docker, Docker Compose, Kafka, Inngest, Git, Vercel CI/CD, Render, OpenTelemetry, Winston, structured logging, health checks

PROJECTS

ShipFlow AI | Next.js, Prisma, PostgreSQL, Inngest, GitHub App, OpenRouter
Live: shipflowai.in | GitHub: github.com/Ayush-Panda-design/ShipFlowAI

- **Problem:** Product teams lose context between ideas, tickets, and pull requests. **Built** a 6-stage delivery workspace (idea to human-approved release) used to unify planning and shipping.
- Ran AI pull request reviews against approved requirements with blocking/non-blocking findings; offloaded 4+ background workflows (reviews, readiness, crons) to Inngest to keep the UI under 2s.
- Integrated GitHub App webhooks and BetterAuth (OAuth + email sessions); deployed with Vercel CI/CD on every push.

Relvion AI | Next.js, PostgreSQL, pgvector, Gemini, MCP, Corsair
Live: relvion-ai.vercel.app | GitHub: github.com/Ayush-Panda-design/Relvion-AI

- **Problem:** Gmail and Calendar are split across tabs with slow search. **Built** a unified AI inbox without migrating user mail out of Google.
- Chose webhook-to-SSE over polling to push new mail in under 1s; used pgvector semantic search as a fallback after operator and cache hits.
- Shipped an MCP agent that drafts, sends, and schedules in one message with validated tool calls.

EdinForm | Next.js, Express, tRPC, Drizzle, PostgreSQL, Redis, Turborepo
Live: edinform.in | GitHub: github.com/Ayush-Panda-design/EdinForm11

- **Problem:** Teams need Typeform-quality forms without enterprise pricing. **Built** a live SaaS form builder on a custom domain with analytics and limits.
- Enforced response caps and expiry server-side across 9 field types; added Redis-backed rate limiting (Upstash) with in-memory fallback for abuse protection.
- Shared Zod validators across a Turborepo monorepo (2 apps, 4 packages) with typed tRPC APIs; split deploy across Vercel and Render.

Votora | React, Express, MongoDB, Socket.io, JWT, Google OAuth
Live: votora-client-jaam.vercel.app | GitHub: github.com/Ayush-Panda-design/Votora-Real-Time-Polling-Platform

- **Problem:** Live events need synchronized polls, not static forms. **Built** realtime rooms with host dashboards and quiz integrity checks.
- Chose Socket.io room broadcasts over SSE because hosts and voters need bidirectional live sync; synced lobby timers across all clients in the same poll room.
- Secured cross-origin auth with JWT httpOnly cookies, Google OAuth, Helmet, CSRF guard, and API rate limits.

OIDC-Compatible Auth Microservice | Node.js, Express, PostgreSQL, OpenTelemetry, Winston, Docker
Production-grade curriculum project | OAuth 2.0 Authorization Code + PKCE + JWKS

- **Problem:** Services need standards-based login instead of ad-hoc JWT logic. **Built** an OIDC-compatible auth service with Authorization Code flow, PKCE, and JWKS rotation.
- Validated issuer, audience, and token expiry on every request; added Winston structured logging, health endpoints, and OpenTelemetry traces for production debugging.
- Containerized the service with Docker Compose for repeatable local deploys and integration testing.

Zanzibar-Style Authorization System | Node.js, PostgreSQL, TypeScript
Curriculum production project | relationship-tuple authorization engine

- **Problem:** RBAC breaks down for nested orgs and shared resources. **Modeled** permissions as relationship tuples with recursive graph traversal.
- Evaluated allow/deny across user, group, and resource hierarchies; compared results against RBAC baselines on multi-level access scenarios.
- Designed schema-first checks before API handlers to keep authorization logic auditable and testable.

EDUCATION

Veer Surendra Sai University of Technology (VSSUT), Burla
B.Tech in Computer Science and Engineering | 2024-2028 | 5th Semester | CGPA: 8.27 | Kendriya Vidyalaya

ENGINEERING PRACTICES

- Applied Kafka in Live Location Tracker (github.com/Ayush-Panda-design/Live-Location-Tracker) with 2 consumer groups to decouple realtime Socket.io broadcasts from MongoDB writes.
- Validates APIs with Zod schemas, TypeScript strict mode, health checks, and structured logs before production deploys.
- Debugged cross-origin auth, webhook retries, and rate-limit edge cases across 4 live apps on Vercel and Render.
- Portfolio: ayushdev-five.vercel.app — 4 production products with case studies, live demos, and resume PDF download.